A Theory of Economic Slack

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CHAPTER 1.

Introducing slack into business cycle research

In 1929, the Great Depression started in the United States. The event spurred a wave of research on business cycles and stabilization policy. Beginning with Keynes, economists naturally concentrated on the depression's most devastating aspect: mass unemployment. A quarter of the labor force was thrown into unemployment, which had terrible consequences—as Steinbeck (1939) described in the *Grapes of Wrath*.

Fifty years later, when the devastation of the depression had faded into memory, modern macroeconomics moved in a different direction: away from unemployment to focus on fluctuations in prices and quantities. But of course, unemployment surges remain a central feature of recessions. They return every once in a while, even when we have forgotten about them.

So this book returns the focus to economic slack—a concept that encompasses unemployment as well as other unsold and idle resources. There are many different forms of slack: people who cannot find a job and remain unemployed; machines left idle in a factory; employed workers left idle on the job; restaurant tables, airplane seats, hotel rooms, conference venues, office buildings that remain vacant; durable goods that cannot be sold and depreciate; nondurable goods that cannot be sold and perish.

The book starts by reviewing essential but little-known facts about slack. It then presents a theory to describe why slack emerges and why it fluctuates. Finally, it uses the theory to explain how monetary and fiscal policy should respond to surges and drops in slack. The book addresses a number of questions that policymakers recurrently face in downturns: How much should interest rates fall when unemployment rises? How large should stimulus spending be? And how early should we start worrying about an upcoming recession?

1.1. Slack in Keynes's General Theory

Perhaps even more than other disciplines, macroeconomics is the product of its time. Trying to make sense of the Great Depression and offer advice to avoid such future calamities, Keynes (1936) wrote the *General Theory of Employment, Interest and Money*. This remains the most influential text in macroeconomics.

The *General Theory* offers a fairly balanced treatment of unemployment and inflation. It discusses inflation and unemployment to a significant extent, with slightly more coverage afforded to unemployment. In the book, inflation and deflation appear on 9 pages; unemployment is mentioned on 32 pages (table 1.1).

Although the *General Theory* does not cover forms of slack beyond unemployment, we know that slack was present in other forms during the Great Depression too: not only was it almost impossible for jobseekers to find jobs, but it was also almost impossible for shops to find customers and for factories to find buyers. As a result, many shops were empty, many factories were idle, and these businesses eventually had to lay their workers off and shut down. As Temin (2000, p. 301) wrote about the Great Depression: "unemployment rose to a peak of 25 percent and stayed above 15 percent for the rest of the 1930s;" at the same time, "there were many idle economic resources in America for a full decade."

This book presents a theoretical framework which shows us that these two phenomena are intrinsically linked. Slack on the product market—the fact that firms cannot find customers because aggregate demand is low—generates slack on the labor market—the fact that workers cannot find jobs because labor demand is low. A key insight of the book is that slack on the product market generates slack on the labor market. Plainly, if firms cannot sell their production, they will not hire workers.

In fact, it is not just that the phenomena of product market slack and labor market slack are linked. It is that they can be modeled entirely symmetrically. The same theory therefore applies to both types of slack, and both markets can be stacked to form a business cycle model with slack.

1.2. Absence of slack in New Keynesian texts

While Keynes was writing during a mass-slack event, by the 1970s, the economic environment had changed, leading to drastic changes in macroeconomic theory. Following the fast economic growth of the US economy in the 1950s and 1960s, and the spike in inflation in the 1970s, the Old Keynesian model—which had formalized Keynes's ideas—was rapidly replaced by the Real Business Cycle model.

TABLE 1.1. Pages mentioning unemployment and inflation in the *General Theory* and New Keynesian texts, according to their indexes

	Keynes (1936)	Gali (2008)	Woodford (2003)
Unemployment	5–9, 13, 15, 16, 22, 109, 121, 128, 190, 235, 251, 274, 275, 278, 289, 334, 335, 346–349, 358, 359, 362–364, 381, 382	188	—
Inflation and related terms	119, 202, 207, 281, 291, 301, 303, 331, 332	1, 4, 11, 21–22, 31–32, 34, 37–38, 47, 49–52, 54–56, 60, 77, 79, 95–99, 103, 105, 107–108, 116, 120, 125–127, 130, 133–135, 137–139, 148, 155–156, 161–164, 168–176, 179, 189	3–5, 13–14, 17, 19–21, 39, 43–44, 46, 49, 90–101, 116–122, 126–138, 159–160, 176–177, 188, 203–205, 212–213, 215, 236, 248, 251, 262, 267, 272, 276–286, 289–295, 301–305, 317, 341, 347, 360–362, 381–382, 396–405, 408–409, 413–416, 418, 437, 440–441, 445–446, 460, 462–463, 467–485, 493, 499–501, 522–527, 544, 559–565, 576–582, 590–604, 615, 619–623, 639–641, 694–695, 714–715, 724–727
Number of unemployment pages	32	1	0
Number of inflation pages	9	60	195
Unemployment share	78%	2%	0%
Inflation share	22%	98%	100%

Besides the term "inflation," Keynes (1936) also features "deflation." In Gali (2008), the terms related to inflation include "inflation dynamics" and "inflation targeting." In Woodford (2003), the terms related to inflation include "inflation inertia," "inflation rates," "inflation-targeting central banks," "inflation-targeting rules," and "deflation." The unemployment and inflation shares are the shares of pages mentioning unemployment and inflation in the total number of pages mentioning either unemployment or inflation.

The Real Business Cycle model was built around Walrasian markets so it had no slack. Because it emphasized efficiency and monetary neutrality, it was not very useful for designing macroeconomic policies, and it was itself replaced by the New Keynesian model in the 1990s.

The New Keynesian model adopted the architecture of the Real Business Cycle model but swapped perfectly competitive markets for monopolistically competitive markets, which allowed researchers to introduce price rigidity and thus break efficiency and monetary neutrality. While the monopolistically competitive model produced rich price dynamics, it did not have room for unemployment or slack either, so the research squarely focused on inflation.

All in all, during the transition from Keynesian macroeconomics to New Keynesian macroeconomics, inflation was retained as a topic of interest but unemployment was lost. This omission of unemployment and slack and concentration on inflation appears clearly in New Keynesian textbooks. The leading New Keynesian texts are *Interest and Prices: Foundations of a Theory of Monetary Policy* by Woodford (2003) and *Monetary Policy, Inflation, and the Business Cycle* by Gali (2008). *Interest and Prices* discusses neither slack nor unemployment. *Monetary Policy, Inflation, and the Business Cycle* does not discuss slack, but unemployment is mentioned on one page at the end of the book, when possible extensions to the New Keynesian model are discussed.

Inflation, by contrast, is ubiquitous in both books (table 1.1). The term "inflation" appears on 97 pages in *Interest and Prices*. Additional related terms, such as "inflation inertia," "inflation-targeting central banks," and "inflation-targeting rules," appear on a further 98 pages. In total, there are 195 pages that cover inflation. In *Monetary Policy, Inflation, and the Business Cycle*, the term "inflation" appears on 52 pages. Related terms such as "inflation dynamics" and "inflation targeting" appear on 8 additional pages. In total 60 pages discuss inflation in that book.

1.3. The cost of ignoring slack: the case of the Great Recession

It turns out that bringing slack back into business cycle research could matter quite a lot. To see this, we return to the start of the Great Recession in early 2008. The unemployment rate was 5.0%.

This was actually below what the New Keynesian literature stipulated. The unemployment target that comes out of the modern literature is the non-accelerating-inflation rate of unemployment (NAIRU): the unemployment rate that keeps inflation stable. Keeping inflation stable has nothing to do with the health of the labor market or the government's mandate of full employment, but because New Keynesian macroeconomics focuses on prices, the NAIRU has naturally become the preeminent unemployment target.[1] Moreover, the New Keynesian model does not natively feature unemployment, so to estimate the NAIRU, unemployment must be introduced somewhat artificially.[2] In any case, the NAIRU was at 5.5% at the time (figure 1.1).

The unemployment rate skyrocketed as the Great Recession unfolded, reaching 9.9% by 2009. The NAIRU also rose to 6.6%, so according to the New Keynesian paradigm, the unemployment rate was 3.3pp too high at the peak of the recession. Furthermore, estimates of the NAIRU are quite uncertain, so it would not be unreasonable to believe that the unemployment rate was only 2.0pp too high at the worst of the Great Recession (Crump et al. 2024, figure 2).

By contrast, this book's theory of slack gives a full-employment rate of unemployment (FERU) below 4.0% during the recession (2008–2009), and below 4.5% at any point during the long recovery from the recession (2010–2015). This means that the unemployment rate was 5.9pp above the FERU in 2009, at the peak of the Great Recession (figure 1.1).[3]

The picture painted by the FERU is therefore much more dire than what the NAIRU suggests. By ignoring slack, the New Keynesian paradigm sets a target for unemployment that is much too lenient: 2.2pp too high in 2008, 2.8pp too high in 2009, and 2.4pp too high in 2010 (figure 1.1). It is not until 2015 that the gap between NAIRU and FERU vanished.

How many jobs would have been saved by targetting the FERU instead of the NAIRU, assuming that policymakers would have tolerated the same unemployment gap under either target? A lot, it turns out. On average between 2008 and 2015, the NAIRU was 1.5pp above the FERU, which translates into 16.3 million worker-years of unemployment.[4] So, by neglecting slack, the New Keynesian paradigm produces unemployment targets that are materially more permissive, allowing for an additional 16 million worker-years of

[1] According to the Council of Economic Advisers (2024, p. 24), "modern economics has generally defined full employment by citing the theoretical concept of the lowest unemployment rate consistent with stable inflation, which is referred to as u^*, ... the non-accelerating inflationary rate of unemployment (termed NAIRU)."

[2] As Crump et al. (2019, p. 166) write: "Although the New Keynesian model—which has become a popular framework for monetary policy analysis and the core structure in many monetary DSGE models—features a forward-looking Phillips curve, it is silent on u^*. Instead of relating inflation to the unemployment gap, it typically relates inflation to the output gap or real marginal costs." An option is to assume that the unemployment rate is the nonparticipation rate, so that fluctuations in unemployment are just negative fluctuations in employment (Hazell et al. 2022, p. 1317). Another option is to assume that the unemployment is the additional amount of labor that would be employed if labor unions did not charge markups; in that case, fluctuations in unemployment reflect fluctuations in the wage markup (Gali 2011, figure 1).

[3] Translating full employment into a number has long been a challenge for policymakers. The 1978 Full Employment and Balanced Growth Act did set an unemployment target of 3% within 5 years of the law, and 4% for young workers (Congress 1978). But these numbers were never taken seriously, in part because policymakers understood that the full-employment unemployment rate could not be a fixed number (DeLong 1996). When Congress asked Alan Greenspan in 1975 what the target for full employment should be, he quipped: "I do not think we should set a target" (Duboff 1977, p. 13).

[4] Each quarter, we compute the gap between NAIRU and FERU and multiply it by the number of workers in the labor force. We then sum these values over the 32 quarters between 2008 and 2015. Dividing by 4 gives 16.3 million worker-years.

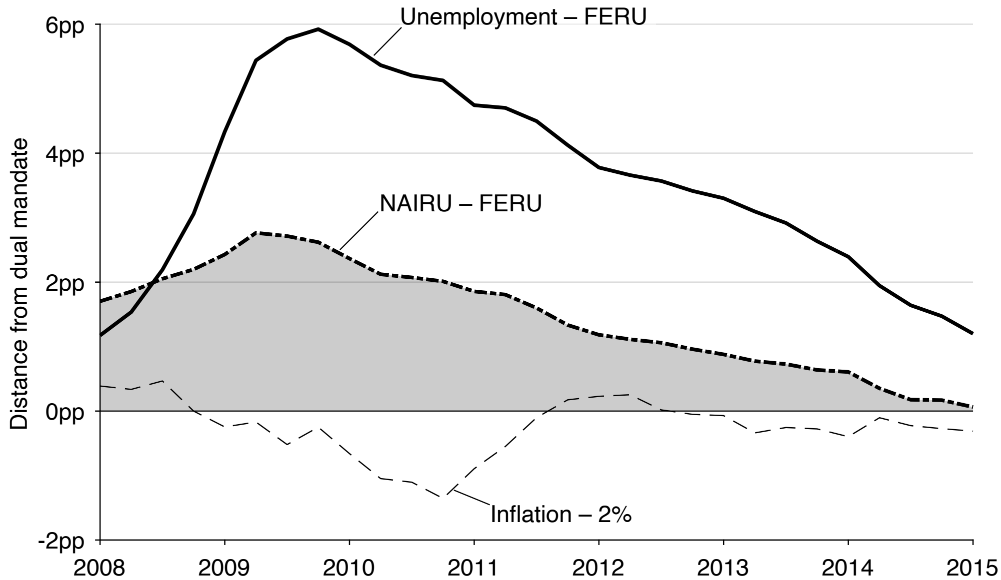

FIGURE 1.1. Costs of ignoring slack during the Great Recession in the United States

The unemployment rate is the quarterly average of the monthly, seasonally adjusted series produced by the Bureau of Labor Statistics (BLS 2025b). The inflation rate is the quarterly average of the monthly, seasonally adjusted, percentage change from a year ago of the consumer price index for all urban consumers, less food and energy, which is produced by the BLS (2025a). The non-accelerating-inflation rate of unemployment (NAIRU) is constructed by Crump et al. (2024, figure 2). The unemployment rates are reported as departures from the full-employment rate of unemployment (FERU), which is described in chapter 12. The inflation rate is reported as departure from the inflation target of 2%.

unemployment during and after the Great Recession. The focus on inflation embodied in the NAIRU is especially odd given that inflation remained 0.3pp below its target of 2% on average between 2008 and 2015 (figure 1.1).

Additionally, the FERU formula developed in this book would have helped dismiss a hypothesis that was popular at the time of the Great Recession and hindered policy response. A substantial number of academic economists and policymakers argued that mismatch was a key cause of high unemployment and therefore that there was not much for the Federal Reserve to do—since it did not have the tool to tackle mismatch.[5] One prominent advocate of that view was Narayana Kocherlakota (2010), who was President of the Federal Reserve Bank of Minneapolis at the time. He argued that the Beveridge curve broke down at the onset of the Great Recession, so that there was not much that the Fed could do about the elevated level of unemployment:

> The relationship [between unemployment and job openings] has completely shattered.... Firms have jobs, but can't find appropriate workers. The workers want to work, but can't find appropriate jobs.... Whatever the source, though,

[5] See Rothstein (2012) for an overview of the debate and related empirical evidence.

> it is hard to see how the Fed can do much to cure this problem. . . . Of course, the key question is: How much of the current unemployment rate is really due to mismatch, as opposed to conditions that the Fed can readily ameliorate? The answer seems to be a lot. . . . Most of the existing unemployment represents mismatch that is not readily amenable to monetary policy.

In essence, Kocherlakota argued that the shift in the Beveridge curve had drastically raised the unemployment rate the Fed should target. On this view, despite the high level of unemployment, the unemployment gap facing the Fed was not very large.

As we will see in the book, the Beveridge curve is indeed central to understanding the workings of the labor market, and is at the heart of our FERU formula. In turn, the FERU formula allows us to quantify how much the Fed's full-employment target increased during the Great Recession. It shows that mismatch only had a small effect on the FERU: yes, the Beveridge curve shifted outward; but no, the rise in the FERU did not account for most of the increase in unemployment. The FERU only increased from 3.8% at the beginning of 2008 to 4.3% in 2010 and 4.5% in 2012. Since the FERU did not increase much, but the unemployment rate increased drastically, the unemployment gap was substantial, peaking at 5.9pp at the end of 2009 (figure 1.1). This means that 5.9pp of unemployment reflected deficient demand and was amenable to monetary policy, and to fiscal policy as well. However, due to the confusion created by the shift in the Beveridge curve and the lack of theory to translate that shift into a new full-employment target, the policy response remained muted, contributing to the slow recovery from the recession.

1.4. Rebalancing business cycle research

Policymakers in many countries have a dual legal mandate: keeping prices stable and maintaining the economy at full employment. Yet the New Keynesian model only speaks about price stability; it says nothing about full employment. The balanced treatment of unemployment and inflation that was a hallmark of Keynesian thought disappeared in the New Keynesian literature, as the focus turned solely to inflation. So the book develops a new business cycle model that provides a more balanced treatment of inflation and unemployment—one that accounts for fluctuations not only in prices, but also in slack, and that can help us think both about price stability and full employment.

Still, why should slack be taken into account to provide an accurate depiction of business cycles and relevant policy advice? The main reasons are that slack varies far more than prices over the business cycle, and slack carries substantial welfare costs. We begin the book in earnest in chapter 2 by making the case that slack is not only prevalent on almost all markets, but also highly countercyclical (low in good times and elevated in bad times). The chapter also argues that economic slack represents a waste of productive resources, and that in addition to its wastefulness, unemployment generates other, large welfare costs to society. Accordingly, a good model of business cycles should feature slack, and good economic policy should avoid periods of elevated slack.

The next question is: How does the book build a business cycle model with slack? The central assumption in the model is that all markets are organized around a matching function that gives the number of trades achieved when a certain number of buyers and sellers look for each other. Chapter 3 introduces the matching function and its properties, and explains why it naturally generates slack.

After chapter 3, the rest of part II develops the theoretical heart of the book: a model of a slackish market. Indeed, to understand why slack matters so much at the macroeconomic level, we must first understand how it operates at the level of individual markets. So, the book starts at the market level: it develops a model of markets with supply, demand, prices, but also slack. It begins in chapter 4 with the most basic slackish market model.

In the slackish market model, prices are set through norms. Because selling is difficult, sellers are glad when they can find a buyer. What this means is that there is a range of prices at which sellers would be willing to sell: the one they had in mind when they placed the good for sale, higher prices of course, but also lower prices that eat up some of their surplus. The same is true for buyers: it is not always easy to find exactly the right good, and buyers are happy when they can purchase what they were looking for, so they would be willing to pay a range of prices. Thus, many price norms can arise in the model. At the same time, the price norm has critical implications for the behavior of the model. Therefore, in chapter 5, we examine how prices are set in the real world to isolate the empirical properties of price norms. Then, in chapter 6, we insert a realistic, somewhat rigid price norm in the slackish market model and explore its properties. After that, we explore several extensions of the basic model: how market capacity is determined (chapter 7), and how the model operates when buyers and sellers enter into long-term customer relationships once they have matched (chapter 8).

We then define market efficiency, which provides a welfare benchmark against which to measure market outcomes (chapter 9). The difficulty in matching buyers and sellers—illustrated by the presence of slack—implies that in reality we are very far from the auction market envisioned by Leon Walras. Many prices might prevail in markets: not just the efficient, market-clearing price that Walras focused on. The implication is that markets generally operate inefficiently. They are sometimes too slack, sometimes too tight, and efficient only in a knife-edge case.

Next, part III applies the slackish market model to the labor market. This is an important application because unemployment is the most socially costly form of slack. There are also good data on labor market slack, which allows us to calibrate the model and assess its quantitative behavior. In chapter 10, we build a slackish labor market model and show that it offers a unified theory of unemployment, merging job rationing and frictional unemployment. In chapter 11, we study the impact of labor market policies in the slackish labor market model: public employment, migration, and unemployment insurance. Last, in chapter 12, we compute the efficient unemployment rate and unemployment gap in the United States over the past century, to assess how far from social efficiency—and therefore full employment—the US labor market has been. We find that the US labor market has been generally inefficiently slack, and especially in recessions.

Having established that the unemployment gap observed in the US labor market is generally positive and sharply countercyclical, we must explain which macroeconomic forces generate such fluctuations and how macroeconomic policies—especially monetary policy—might stabilize them. This is what parts IV and V aim to do.

Part IV scales up the model from a slackish market to a slackish economy. We start with a basic one-market economy in chapter 13 to understand the core properties of slackish business cycles. With this basic model, we see that aggregate demand shocks can generate countercyclical fluctuations in the unemployment gap, and that monetary and fiscal policy can counteract such fluctuations. In chapter 14, we introduce directed search into the business cycle model to produce a Phillips curve that links inflation to slack. In chapter 15, we extend the model to a more realistic economy with both labor and product markets. The two-market economy allows us to understand how slack percolates from the product market to the labor market.

With a complete business cycle model in hand, we turn in part V to the policy questions posed by the unemployment-gap evidence: How should monetary and fiscal policy be conducted when the economy is not at full employment? We derive formulas for the optimal conduct of monetary policy (chapter 16) and fiscal policy (chapter 17) based on the unemployment gap. The part closes by showing how slack data can be used to detect recessions early and therefore trigger early policy responses (chapter 18).

Part VI concludes by summarizing the main arguments and implications of the slackish framework (chapter 19). It then outlines how the theory developed in the book could be fruitfully applied to countries besides the United States, particularly developing countries, where slack is widespread (chapter 20). It closes by situating the book's slackish business cycle model within the macroeconomic landscape—drawing connections to the Old Keynesian, Real Business Cycle, and New Keynesian models (chapter 21).

So, who is this book for? I have aimed to write the book as a text in quantitative social science. As such, it requires some basic mathematical training, at roughly the undergraduate level, as well as a curiosity about the social sciences, since it draws at various points on evidence and theories from psychology, sociology, public health, and economic history. Anyone with this background should be able to read the book from beginning to end, absorb a coherent picture of how the economy fluctuates over the business cycle, and later apply the book's theory to their own research.

Although some sections of the book are necessarily mathematical, I hope the ideas developed here will speak to general readers interested in how modern economies function and sometimes falter. I have aimed to make the data, models, and findings accessible to anyone who wishes to follow the central argument without absorbing every mathematical detail. To make this possible, the book relies throughout on theoretical diagrams and empirical plots so that all arguments can be understood visually. Modern macroeconomics has abandoned models that can be represented and analyzed diagrammatically; but a theory that can be understood, manipulated, and extended through visual reasoning might have certain advantages over a theory that can only be understood through numerical solution of large dynamical systems.

It is true that this book diverges conceptually and methodologically from the traditional macroeconomic approach. Professional macroeconomists might find the material unfamiliar and difficult to engage with. When I presented the concept of FERU at a policy conference in Washington, DC, in 2023, Christina Romer criticized our effort to "divorce the concept of full employment from stable prices" on the grounds that "a concept of full employment that isn't consistent with stable inflation is not a sensible goal for policy".[6] Romer is one of the most accomplished scholars of the Great Depression; an expert on business cycles; and a leading authority on monetary and stabilization policy.[7] Furthermore, she understands how policy decisions are made in the real world: she chaired the Council of Economic Advisers at the time of the Great Recession, hired by the White House to guide the US economy out of the crisis. Her response reflected a common view—full employment should be defined through price stability—and revealed how inflation-centric current policy thinking is. Nevertheless, I hope that the book might entice macroeconomists to make a bit more room for slack in business cycle research.

[6]The quote is from the discussion that follows our paper on the FERU (Michaillat and Saez 2024, p. 422).

[7]See for instance Romer (1993), Romer (1999), and Romer and Romer (2023). Her expertise is widely recognized: she was codirector of the NBER Monetary Economics Program for two decades and is part of the NBER Business Cycle Dating Committee.

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